# Formula Slug - Power Electronics Onboarding

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October 2026

## 1 Introduction

Hi, my name is Joseph and this project has been super fun. I've already learned quite a bit from it

## 2 Calculations

The converter is designed for 12V input, 5V 2A output. Using the equation:

$$V_{out} = DV_{in}$$

We find that we want a duty cycle of $D = 0.4167$, where the switch is on approximately 41.67% of the time.

I chose a switching frequency of $f_s = 700$kHz, which is above the 500kHz specified in the document

To calculate the component values, I used the formulas:

$$L = \frac{(V_{in} - V_{out}) \cdot D}{\Delta I_L \cdot f_s}; \quad C = \frac{\Delta I_L}{8 f_s \cdot \Delta V}$$

# 3 Sourcing

| Component | Cost | Link | Units | Size |
|---|---|---|---|---|
| Buck IC | $2.14 | TPS5405DR | 1 | |
| Inductor | $1.54 | PF0560.702NL | 1 | 10.4 x 10.3mm |
| Capacitor | | | | |
| Total Cost | | | | |